

Initial Project Planning Template

Date	10 July 2026
Team ID	XXXXXX
Project Name	Rising Waters: A Machine Learning Approach to Flood Prediction
Maximum Marks	5 Marks

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members	Sprint Start Date	Sprint End Date (Planned)
Sprint-1	Dataset Collection	USN-1	As a developer, I want to collect historical rainfall and flood datasets so that the machine learning model can be trained accurately.	3	High	Harika Velugubantla	10-07-2026	12-07-2026
Sprint-1	Data Analysis	USN-2	As a developer, I want to analyze the dataset using statistical summaries and visualizations to understand rainfall and flood patterns.	3	High	Harika Velugubantla	10-07-2026	13-07-2026
Sprint-2	Data Preprocessing	USN-3	As a developer, I want to clean, preprocess, and prepare the dataset for machine learning model training.	5		Harika Velugubantla	14-07-2026	16-07-2026

Sprint-2	Model Development	USN-4	As a developer, I want to train and compare Random Forest, Decision Tree, KNN, and XGBoost models to identify the best-performing model.	8	High	Harika Velugubantla	14-07-2026	18-07-2026
Sprint-3	Model Evaluation	USN-5	As a developer, I want to evaluate the trained models using accuracy, confusion matrix, and classification report to select the optimal model.	3	High	Harika Velugubantla	19-07-2026	20-07-2026
Sprint-3	Model Saving	USN-6	As a developer, I want to save the trained machine learning model and preprocessing files for future predictions.	2	High	Harika Velugubantla	19-07-2026	20-07-2026
Sprint-4	Flask Web Application	USN-7	As a user, I want to enter rainfall and weather parameters so that I can receive an instant flood prediction.	5	High	Harika Velugubantla	21-07-2026	24-07-2026
Sprint-4	Prediction Result Interface	USN-8	As a user, I want to view flood prediction results through a simple and interactive web interface.	3	High	Harika Velugubantla	21-07-2026	24-07-2026
Sprint-5	Testing & Deployment	USN-9	As a developer, I want to test the complete system and deploy the application so that users can access flood predictions online.	5	High	Harika Velugubantla	25-07-2026	28-07-2026